Self-Supervised Feature Learning for Multimodal Remote Sensing Image Land Cover Classification

Zhixiang Xue, Xuchu Yu, Anzhu Yu, Bing Liu, Pengqiang Zhang, and Shentong Wu

***Abstract*— Deep learning models have shown great potential in remote sensing (RS) image processing and analysis. Nevertheless, there are insufficient labeled samples to train deep networks, which seriously affects the performance of these models. To resolve this contradiction, we propose a generative self-supervised feature learning (S2FL) architecture for multimodal RS image land cover classification. Specifically, multiple complementary observed views are constructed from multimodal RS images, which are employed for following generative self-supervised learning (SSL). The proposed S2FL architecture is capable of extracting high-level meaningful feature representations from multiview data, and this process does not require any labeled information, providing a feasible solution to relieve the urgent need for annotated samples. The learned features are normalized and merged with corresponding spectral information to further improve the discriminative capability of feature representations, and we utilize these fused features for land cover classification. Compared with the existing supervised, semi-supervised, and self-supervised approaches, the proposed generative self-supervised model achieves superior performance in terms of feature learning and land cover classification, especially in the small sample classification case.**

***Index Terms*— Feature learning, generative self-supervised learning (SSL), land cover classification, multimodal data, remote sensing (RS), transformer structure.**

I. Introduction

THE recognition and classification of surface objects are an important part of human understanding of the earth's environment. Remote sensing (RS) technology obtains the spectrum properties of land cover objects in a contact-free means. The imaging spectrometer records detailed spectrum characteristics of ground objects; thus, the acquired hyperspectral images (HSIs) contain abundant spectral diagnostic information for the precise recognition of land cover objects. The hyperspectral RS has been successfully applied in crop precise identification [1], tree species identification [2], heterogeneous crops identification [3], and coastal wetlands mapping [4]. Apart from hyperspectral data, very high-resolution (VHR) images, synthetic aperture radar (SAR), and light detection and ranging (LiDAR) data contain rich spatial distribution and precise object shape information of ground objects [5]. Multimodal RS images have been applied in land cover classification [6], crop classification [7], and volcano mapping [8]. Therefore, it is a promising paradigm to utilize rich and complementary information contained in multimodal RS data for land cover collaborate classification.

The supervised classification methods are the most commonly utilized classification scheme in the land cover classification field. Several typical deep neural networks have been utilized to construct end-to-end classification models. Convolutional neural networks (CNNs) can extract locally invariant features, and 1-D CNN, 2-D CNN, and 3-D CNN learn local spectral–spatial information for classification [9]–[11]. The capsule networks (CapsNet) encode high-level discriminative features into vectors for HSI classification [12]. As a sequence model, the recurrent neural network (RNN) treats HSI as sequence data to extract sequence information in spectral space [13]. To explore the long-range dependencies between pixels, the transformers specific to RS data are investigated to further improve classification performance [14], [15]. In addition to the abovementioned model-driven approaches, some machine learning strategies are also developed for land cover classification, such as attention mechanism [16], ensemble learning [17], multiscale learning strategy [18], and so on. As data-driven paradigms, these supervised deep learning models require numerous annotated samples for training. However, it is laborious to annotate high-quality samples in RS field; thus, insufficient labeled samples have been earmarked as a significant bottleneck for the supervised classification paradigm.

Over the last few years, intense research has proposed a plethora of approaches aiming at solving the small sample classification problem. The semi-supervised learning paradigm utilizes both labeled samples and unlabeled ones for land cover classification, and the graph-based learning is a typical semi-supervised pattern for RS data [19]. The graph convolutional networks (GCNs) utilize both labeled and unlabeled samples to construct graph structures; then, deep graph features are extracted for classification [20], [21]. As a data augmentation model, generative adversarial network (GAN) employs annotated as well as generated samples for semi-supervised classification [22], [23]. Because of the complex characteristics of RS data, these semi-supervised models have difficulties in generating realistic samples and constructing graph structures. Another research direction is the weakly

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supervised learning, in which Siamese networks are employed to extract high-dimensional nonlinear spatial–spectral features from a limited number of labeled samples [24]. The few-shot learning scheme trains a feature extractor from source domain and applied directly to target domain for feature extraction, and these extracted features are employed for classification with a few labeled samples [25]. The transfer learning mainly consists of pretraining and fine-tuning stages, in which low-level as well as midlevel features are learned from the source dataset in the pretraining stage, and the network is tweaked with a few labeled samples for the target classification task [26]. To solve the domain shift problem, the domain adaptation aims to minimize the domain gap between the source and target domains, attempting to reduce the difference in the statistical distributions [27]. Despite the remarkable performance achieved by these models, current classification schemes cannot effectively solve the most outstanding problem; that is, there are numerous unlabeled images but a limited number of annotated samples.

As a special type of representation learning, the self-supervised learning (SSL) aims to learn meaningful feature representations from unlabeled data by utilizing the inherent attributes and relationships. Because it is time-consuming and labor-intensive to annotate high-quality samples in real situation, learning good representation provides a promising solution to relieve the urgent need for manually annotated data. The general paradigm of SSL is to pretrain the model using the inherent characteristics of unlabeled data as supervision signal; then, the trained model is employed for unsupervised feature extraction, and the learned meaningful feature representations are more conductive to the following specific task. Some intrinsic attributes are exploited for hyperspectral data SSL, such as the low-rank structure [28], multilevel features constraints [29], spectral–spatial consistency [30], intracluster similarity, and intercluster dissimilarity [31]. The spatial–spectral similarity and adaptive distillation technology are also utilized for SSL of hyperspectral data [32]. In addition, the encoder–decoder structures are investigated for HSI self-supervised feature learning (S2FL) [33], [34]. The mutual information constraint contained in polarimetric SAR (PolSAR) images is also employed for SSL [35].

However, current SSL models are investigated for single modal RS data, which cannot fully utilize rich and complementary information contained in multimodal data for feature learning. The CNN architecture is adopted in the existing SSL models; thus, these networks have a limited range of receptive fields. In this article, we investigate a generative SSL architecture for multimodal RS images feature learning and land cover classification, which can not only construct multiple complementary views from multimodal RS data, but also learn semantically meaningful feature representations using generative SSL without any labeled information. The main contributions are summarized as follows.

1) We bring forward to construct multiple complementary views from multimodal RS images, which generally have rich and complementary information. Also, these constructed multiple views are employed for S2FL.
2) The generative SSL architecture is investigated for multimodal RS data feature learning, which consists of the asymmetric encoder and decoder structure, and the transformer structure is employed as backbone for encoder as well as decoder. This scheme is capable of learning key feature representations characterizing the original multimodal RS data.
3) To further improve the discriminative capability of features, we combine the learned spatial features with corresponding spectral information, so that the integrated feature representations are more conductive to the following land cover classification.
4) We qualitatively and quantitatively evaluate the performance of the proposed S2FL model on four benchmark RS datasets, which achieves superior performance in terms of feature learning and land cover classification, especially in small sample classification situation.

The remainder of this article is organized as follows. Related work regarding SSL for RS data is introduced in Section II. Section III elaborates on the generative self-supervised network for multimodal RS image land cover classification. The experimental results are analyzed in Section IV. In Section V, this article is discussed and summarized.

II. Related Work

As a promising machine learning paradigm, SSL leverages the inherent characteristics of input data to learn meaningful feature representations, which provides a feasible solution to alleviate the high dependency on annotated samples. SSL has shown great potential in natural language processing (NLP) and computer vision (CV) fields, and there are some approaches that have been investigated for RS image S2FL in recent years. There are two representative SSL schemes, i.e., contrastive and generative SSL approaches.

A. Contrastive SSL

By comparing the augmented views derived from samples, the contrastive learning aims to learn both invariant and discriminative feature representations from unlabeled samples. The cores of contrastive learning are data augmentation and the construction of loss function. The data augmentation operation is to generate multiple views from the same sample and consider them as similar pairs, while views derived from different samples are treated as dissimilar ones. The contrastive loss function makes similar pairs stay close while separating dissimilar ones in the latent feature space [36].

Recently, there are several self-supervised models investigated for RS image processing and analysis. Two augmented views are constructed from hyperspectral data by different augmentation methods, which are employed for contrastive learning in the pretraining stage, and learned features are used for land cover classification in the fine-tuning phase [37]–[39]. In addition to the contrastive learning conducted in the spatial domain, the cross-domain contrastive learning frameworks are also investigated to learn high-level domain-invariant features from hyperspectral data [40], [41]. Because of its excellent feature learning and generalization properties, the contrastive learning architecture has also been investigated for RS images scene classification [42], semantic segmentation [43], PolSAR land cover classification [44], SAR-optical data fusion [45],

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and change detection [46]. These models are developed for single modal data feature learning, in which the CNN is employed as the backbone.

B. Generative SSL

The generative SSL achieves the goal of feature learning by restoring the manually corrupted data. The motivation behind this is that once the model can restore the original data, it means that the model has learned key features characterizing the original data. As a classical method of generative SSL, the autoencoder maps input data to latent features using the encoder structure; then, these feature are reconstructed into the input signal by the decoder structure [47]. The generative self-supervised pretraining schemes are highly successful in the NLP field. These methods generally mask a portion of the input sequence and try to predict the missing content, which can be trained on large amounts of unlabeled data. However, the application of generative SSL in vision field lags behind. The main reason is that languages are highly semantic while images are natural signals with heavy spatial redundancy; thus, it is difficult for generative SSL models to learn high-level key information from images.

By dividing the input image into patches, the vision transformer (ViT) can obtain the global receptive field of the entire image [48]. Setting a high masking ratio, the recent generative SSL models employ the transformer structure as backbone to learn high-level meaningful information from images and videos [49], [50]. In the RS image processing field, there are multimodal RS images within the same observed scenes, and these heterogeneous images generally have rich and complementary information. However, current generative SSL methods cannot make full use of these heterogeneous data to learn high-level meaningful feature representations.

III. Methodology

In this section, we elaborate on the proposed generative self-supervised pretraining and few-shot land cover classification method for multimodal RS data. This approach mainly contains two stages, i.e., generative self-supervised pretraining and few-shot land cover classification. In the pretraining procedure, constructed from multimodal RS data, local multiview observed images are divided into image patches, which are masked randomly, and unmasked patches are embedded for the deep transformer encoder to learn high-level feature representations; thereafter, the shallow decoder reconstructs the learned features as well as masked patches to original images in the pixel space. After the self-supervised pretraining process, the samples of the same class aggregate into clusters in the learned feature space, and samples of different categories are separated away. In the land cover classification phase, the learned spatial features are normalized and combined with corresponding spectral information, which are employed as input of the lightweight support vector machine (SVM) classifier for classification.

A. Self-Supervised Pretraining

The investigated generative S2FL approach adopts asymmetric encoder–decoder architecture, in which the encoder maps the unmasked multimodal image patches into latent space, and the decoder reconstructs original image using learned features as well as masked image patches. The transformer structure is employed as a backbone for the encoder and the decoder. The schematic architecture of the self-supervised pretraining model is depicted in Fig. 1.

Because multimodal RS images contain rich and complementary information, we employ multimodal RS data for S2FL. Due to the redundant information between spectral bands of hyperspectral data, employing all spectral bands complicate the training process of SSL, and the feature learning ability is poor. We propose the multiview learning strategy to construct multiple observed views from multimodal RS data. The HSIs are divided into several groups, and we adopt the principal component analysis (PCA) operation to extract first three principal components in each group. Because the VHR image contains specific spectral information, we directly employ it as individual observed view. For LiDAR data in the same observed scenes, we employ the invariant attribute profiles (IAPs) [51] to extract three-band profiles from LiDAR-derived digital surface model (DSM) as a single view. Then, these constructed views are normalized and stacked as multiview data for S2FL. Centering around each RS image pixel, we crop local image patches of size $H \times W$ as processing unit for following generative feature learning.

Let $\boldsymbol{x}_n \in \mathbb{R}^{H\times W\times B}$ represent multiview data sample, in which H, W, and B denote the height, width, and depth, respectively. The sample is divided into several regular nonoverlapping image patches. A subset of image patches is randomly masked, and the unmasked patches are embedded with a trainable linear projection to obtain the input embeddings, denoted as T_v. Each masked token is represented by a trainable vector, which only indicates the presence of patch to be predicted, denoted as T_m.

As shown in Fig. 1, the encoder consists of several transformer layers, and visible tokens T_v are utilized as input to the encoder. The class token is also added to the input tokens, which serves as feature representations of input sequence at the output of the encoder. The relative position encoding operation is also employed to provide location information. After the position embedding operation, the embedded features are input into the transformer-based encoder for feature learning, which can be formulated as follows:

$$T_e = \text{Encoder}(T_v). \tag{1}$$

The transformer structure consists of alternating multihead self-attention (MHSA) and multilayer perceptron (MLP) blocks, and the illustration of transformer structure is shown in Fig. 2. The attention mechanism can adaptively model the long-range dependencies of sequence tokens, and a single attention function only establishes one kind of dependency. It is a fact that there are complex multiple dependencies between sequence tokens, and the MHSA aims to learn multiple dependencies from different representation subspaces. Specifically, d_{model}-dimensional keys, values, and queries are linearly divided into h groups, and the attention function is performed in all groups in parallel, yielding d_{model}/h-dimensional output values, which are concatenated

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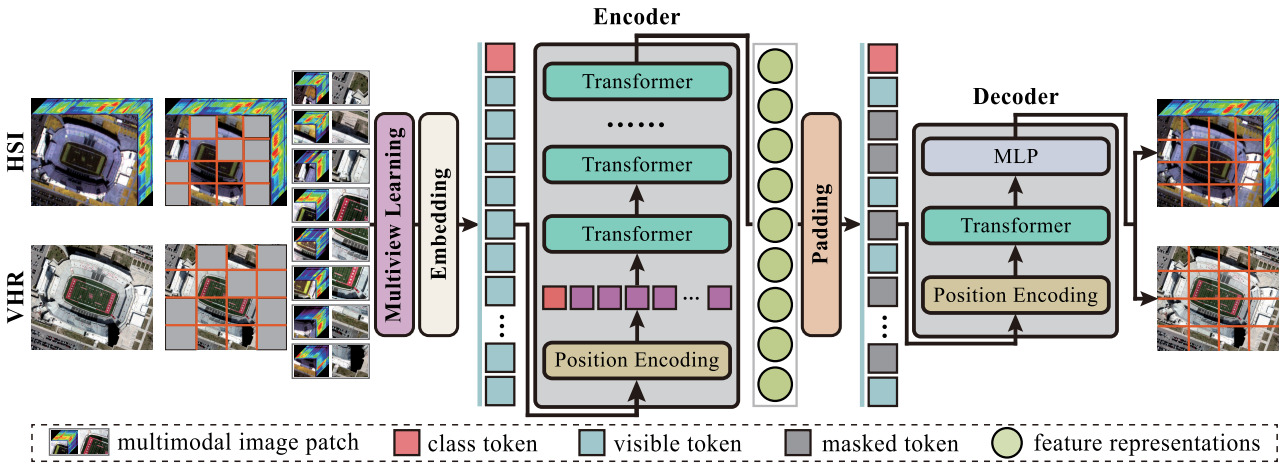

Fig. 1. Illustration of the investigated generative self-supervised pretraining architecture.

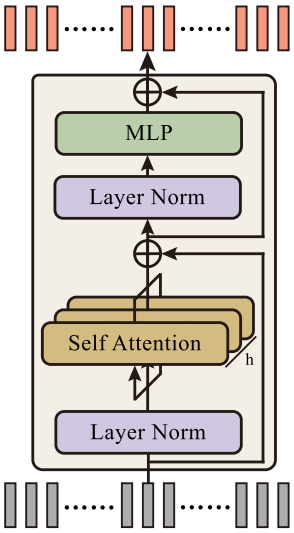

Fig. 2. Illustration of the transformer structure. The transformer mainly consists of MHSA and MLP structures, and layer normalization as well as residual connections are also included.

and projected. The MHSA can be formulated as follows:

$$\text{MultiHead}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Concat}(\boldsymbol{y}_1, \ldots, \boldsymbol{y}_h)\boldsymbol{W}^O$$
$$\boldsymbol{y}_i = \text{softmax}\left(\frac{\boldsymbol{Q}_i \boldsymbol{K}_i^T}{\sqrt{d_k}}\right)\boldsymbol{V}_i \tag{2}$$

where h is the number of attention heads, $\boldsymbol{W}^O$ denotes learnable projection matrix, and $d_k = d_{\text{model}}/h$ represents the feature dimension of each head. $\boldsymbol{Q}_i = \boldsymbol{X}_i \boldsymbol{W}_i^q$, $\boldsymbol{K}_i = \boldsymbol{X}_i \boldsymbol{W}_i^k$, and $\boldsymbol{V}_i = \boldsymbol{W}_i^v$ represent the key, query, and value, respectively, in which $\boldsymbol{W}_i^q$, $\boldsymbol{W}_i^k$, and $\boldsymbol{W}_i^v$ are parameter matrices, and $\boldsymbol{X}_i$ denotes the feature matrix for the ith head.

The decoder is similar to the encoder but containing fewer transformer layers, and the input for the decoder is the encoded visible patches T_e as well as masked tokens T_m, as shown in Fig. 1. The padding operation adds the masked tokens to the input sequence according to original position. We also perform the position encoding operation to provide relative location information in the reconstruction task. Also, the decoder can be formulated as follows:

$$T_d = \text{Decoder}(T_e, T_m). \tag{3}$$

As the last layer of decoder, the prediction head aims to reconstruct image patches in the pixel space. We simply employ a fully connected layer as the prediction head. Taking the output T_d from the decoder, the prediction head projects it to a vector, which has the same dimension as the input sample. Then, followed by a reshape operation, predicted patches P are obtained, which can be expressed as follows:

$$P = \text{Reshape}(FC(T_d)). \tag{4}$$

After obtaining the reconstructed patches, we normalize the original and reconstructed patches separately, and the reconstruction target computes the mean squared error (MSE) between the reconstructed and original patches. Note that the reconstruction loss is only computed on masked patches and corresponding reconstructed patches.

Using this asymmetric encoder–decoder structure paired with large mask ratio, the deep encoder maps a subset of patches into latent feature space; then, the shallow decoder reconstructs the original images using learned features and masked tokens. In the pretraining procedure, the self-supervised scheme is trained with a large amount of unlabeled data. After the pretraining process, the decoder is discarded, and the trained encoder is employed as unsupervised feature extractor, and extracted features serve as feature representations of the original data.

B. Land Cover Classification

After the S2FL, the learned key features are more conductive to the following classification task, and the schematic architecture of land cover classification is shown in Fig. 3. In the supervised classification stage, the trained encoder serves as unsupervised feature extractor. Also, all samples of multimodal RS dataset pass through this feature extractor to output corresponding feature representations, since these features are extracted from local RS images, which mainly represent local spatial information. To further enhance the discriminative capability of features, the learned features are normalized and combined with corresponding spectral information as the final feature representations. After the above-mentioned feature learning, samples of the same category will

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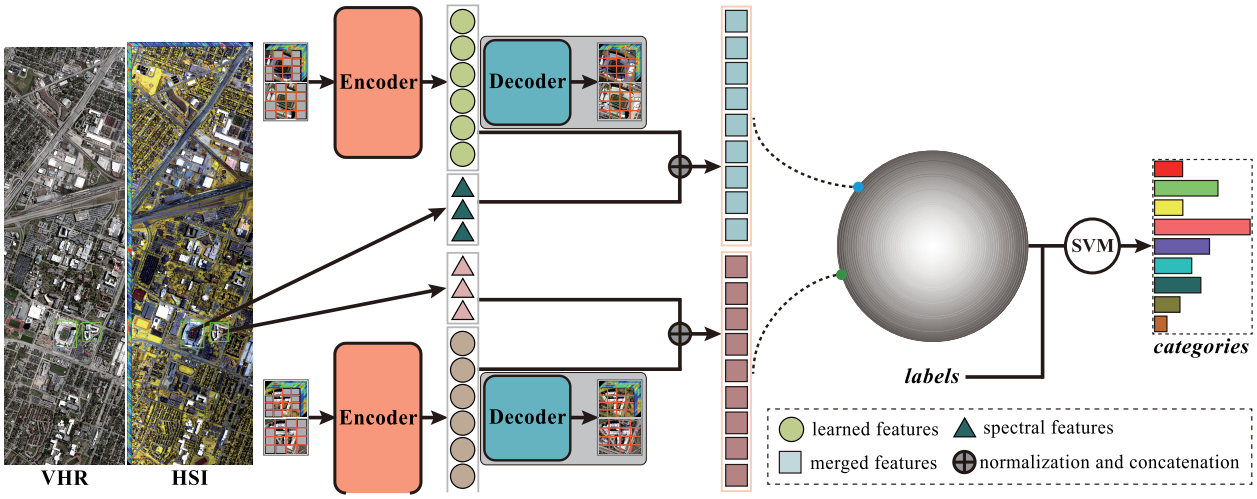

Fig. 3. Schematic architecture of generative S2FL for multimodal RS images land cover classification. The input of this framework is multimodal RS data, and multiple observed views are constructed from multimodal RS images for feature extraction. The extracted features are combined with corresponding spectral information; then, the SVM is utilized for land cover classification.

gather together in the latent feature space, and samples of different classes will be separated away. Thus far, the SVM classifies the combined feature representations with very few labeled samples.

IV. Experiments and Analysis

In this section, comparative experiments are reported to certify the effectiveness and superiority of the devised S2FL method. The proposed algorithm is realized within the PyTorch library, and other methods are implemented in the original experimental environments. All classification experiments are carried out on a workstation equipped with two NVIDIA GeForce RTX 3090 graphics cards, and this workstation has a memory of 192G.

A. Datasets Description

In this research, four benchmark RS datasets are employed to conduct the generative S2FL and land cover classification experiments.

The first dataset is the Salinas Valley hyperspectral data, which consists of 204 spectral bands with a spatial size of 512 × 217 pixels. The second dataset is the Pavia University hyperspectral data, which consists of 103 bands with a spatial size of 610 × 340 pixels. The third dataset is Trento, a hyperspectral and LiDAR dataset, and the hyperspectral imagery consists of 63 spectral bands with a spatial size of 166 × 600 pixels. The fourth dataset is the Houston 2018, a hyperspectral and VHR dataset, in which the HSI consists of 48 spectral bands with a spatial size of 601 × 2384 pixels [52]. The visualization of these datasets and thematic distribution map are shown in Figs. 4–7.

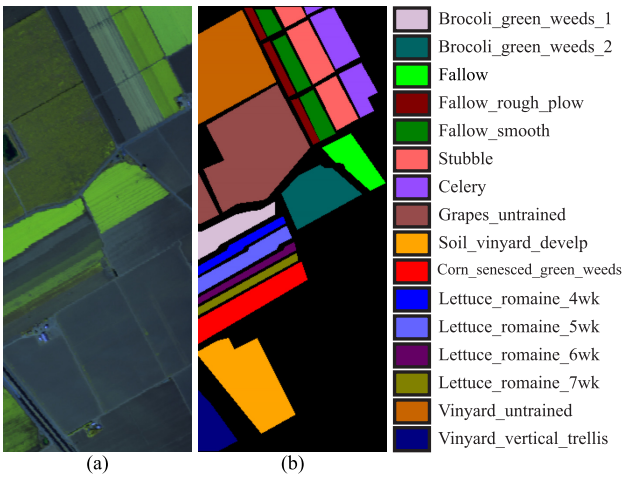

Fig. 4. Visualization for Salinas Valley scene. (a) Simulated-color image of hyperspectral data. (b) Thematic distribution map.

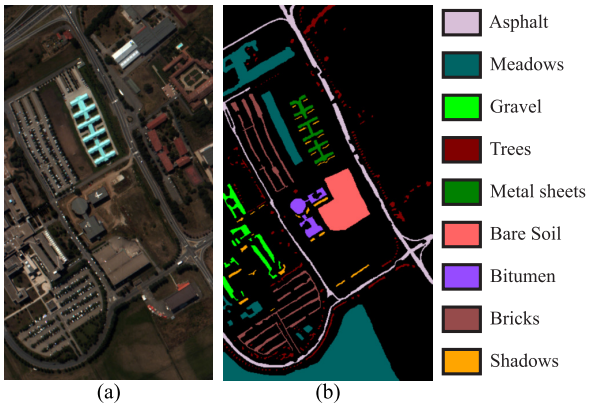

Fig. 5. Visualization for Pavia University scene. (a) Simulated-color image of hyperspectral data. (b) Thematic distribution map.

B. Parameter Setting and Analysis

In the pretraining procedure, the Adam optimizer is used to optimize the S2FL model. The learning rate is set to be $1e-4$, and the batch size is 256. The embedding dimension of the transformer structure is 512, and the number of heads is set as 8. We set the number of epochs for the Houston 2018 dataset as 60, and the number of epochs for the other three datasets is 300. The regularization parameter C and kernel parameter γ in SVM are optimized using the cross-validation strategy.

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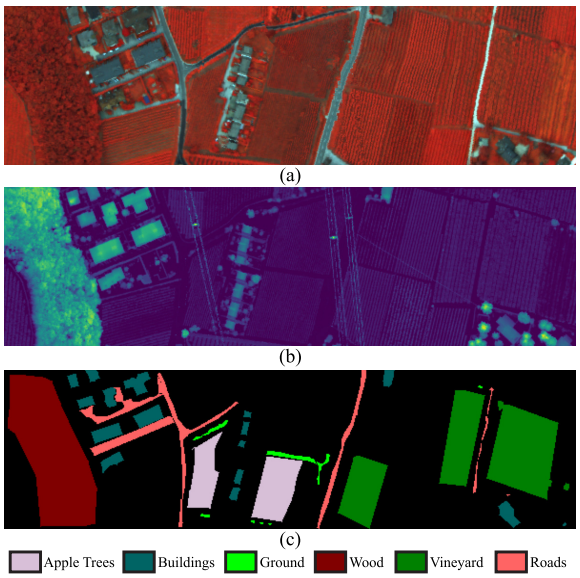

Fig. 6. Visualization for Trento scene. (a) Simulated-color image of hyperspectral data. (b) Grayscale image of DSM. (c) Thematic distribution map.

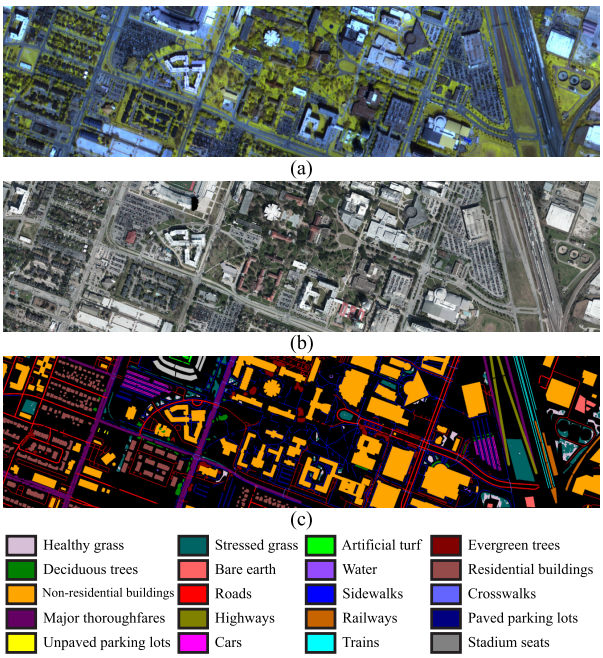

Fig. 7. Visualization for Houston 2018 scene. (a) Simulated-color image of hyperspectral data. (b) VHR image. (c) Thematic distribution map.

Some hyperparameters in the proposed S2FL approach affect the feature learning to a large extent, which, in turn, influence the land cover classification performance. Thus, we analyze the influences of these hyperparameters on classification performance.

1) Model Structure: The network structure of S2FL model affects the number of parameters and feature extraction capability, then affecting following classification performance. The asymmetric encoder–decoder structure is adopted in the S2FL model, in which the encoder and the decoder mainly consist of transformer structures. We set different transformer layers for encoder and decoder, and candidate sets are set to be $\{6, 8, 10, 12\}$ and $\{2, 3, 4\}$, respectively. The classification results with different model structures are reported in Fig. 8. From Fig. 8, we can observe that deep encoder paired with shallow decoder could learn more discriminative features, resulting in higher classification accuracies. Also, reasonable network structures for four RS datasets are $\{10, 2\}$, $\{8, 2\}$, $\{10, 2\}$, and $\{10, 2\}$, respectively.

2) Mask Ratio: The input of the encoder is unmasked image patches, which enables the encoder to learn key features characterizing original data. This parameter affects the number of input image patches, which, in turn, influences the feature learning capability. The overall classification accuracies with different mask ratios on four benchmark datasets are reported in Fig. 9. It can be observed that this parameter has large impact on the classification performance, and optimal mask rates corresponding to four benchmark datasets are 0.5, 0.4, 0.5, and 0.5, respectively.

3) Image Dimension: Since we construct multiview data from RS images for SSL, we analyze the model's feature learning capability with different numbers of observed views. For LiDAR data, we employ the IAPs operation to extract three-band profiles from LiDAR-derived DSM as a single view, and the VHR image is employed as a single view. We employ the PCA operation to extract multiple observed views from hyperspectral data. The candidate set of constructed views from hyperspectral data is set as $\{1, 2, 3, 4\}$, and the classification results with different views are reported in Table I. Because the model could only extract limited information from single observed view, its classification accuracy is generally low. The model could extract more discriminative features from a moderate number of observed views, and the optimal views for four datasets are 2, 2, 3, and 2, respectively.

4) Patch Size: The input of the proposed generative SSL model is local multiview data, and the patch size affects the receptive field of the transformer structure. The change trends of overall accuracies using different spatial sizes are shown in Fig. 10. Larger image patches allow the model to obtain larger receptive fields, but irrelevant redundant information also affects the classification performance. The spatial resolution of RS images affects the optimal spatial size, and datasets with higher spatial resolution correspond to larger patch sizes. According to the experimental results, reasonable patch sizes for four benchmark datasets are 36, 36, 30, and 42, respectively.

5) Labeled Samples: Since the proposed SSL model extracts features for the classification task, we analyze the classification performance with different numbers of labeled samples, and corresponding overall accuracies are reported in Fig. 11. It can be observed from Fig. 11 that the proposed approach achieves high classification accuracy even with a small number of labeled samples, e.g., three, five, and ten labeled samples per class, which indicates that this SSL model could effectively extract discriminative features for classification task.

6) Model Complexity Analysis: The model complexity is another important indicator for evaluating neural network models, and we conduct quantitative analysis of model complexity in this research. Model complexity mainly includes space complexity and time complexity; thus, we utilize

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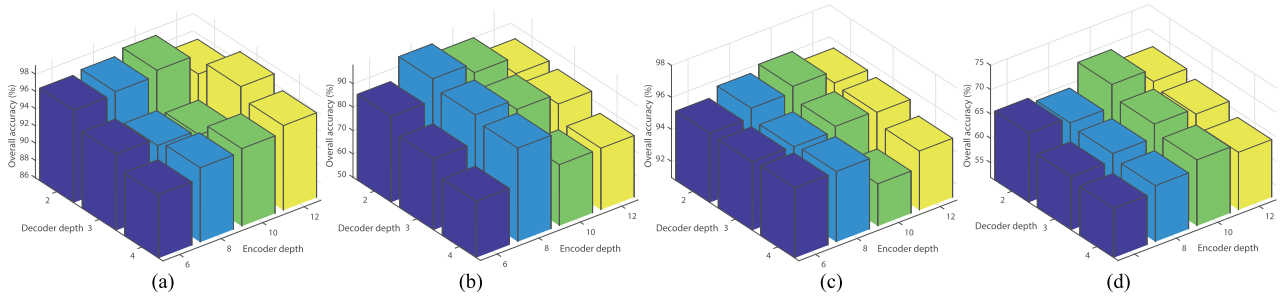

Fig. 8. Sensitivity analysis of model structure on four benchmark datasets. (a) Salinas Valley. (b) Pavia University. (c) Trento. (d) Houston 2018.

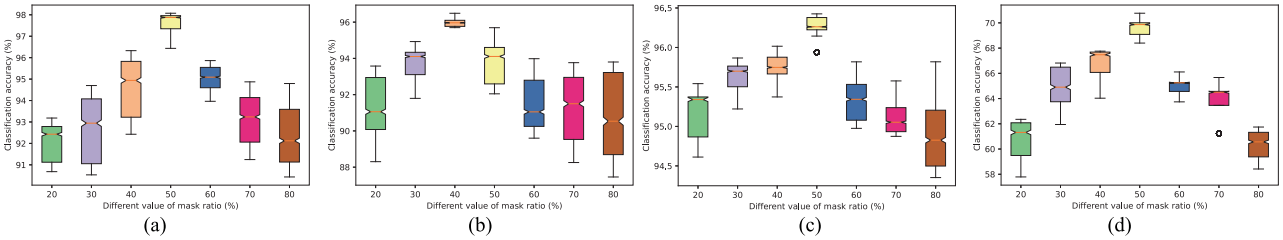

Fig. 9. Sensitivity analysis of mask ratio on four benchmark datasets. (a) Salinas Valley. (b) Pavia University. (c) Trento. (d) Houston 2018.

TABLE I

CLASSIFICATION RESULTS WITH DIFFERENT CONSTRUCTED VIEWS ON FOUR BENCHMARK DATASETS

Views	Salinas Valley			Pavia University			Trento			Houston 2018		
	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
1	90.45	93.64	89.40	87.11	87.72	83.28	94.12	88.93	92.16	61.62	49.97	54.87
	0.46	0.50	0.51	0.34	0.20	0.43	0.38	0.75	0.51	0.09	0.11	0.10
2	97.66	97.41	97.40	96.00	94.93	94.72	95.61	93.08	94.15	69.63	63.15	63.03
	0.17	0.16	0.22	0.24	0.23	0.21	0.24	0.39	0.33	0.13	0.22	0.14
3	95.64	95.63	95.15	90.68	91.82	87.66	96.26	93.26	95.00	66.58	59.28	60.02
	0.34	0.33	0.38	0.27	0.18	0.35	0.13	0.21	0.17	0.08	0.11	0.13
4	96.95	97.25	96.60	84.05	87.70	80.03	94.27	91.73	92.36	63.40	56.93	56.61
	0.45	0.27	0.50	0.15	0.09	0.18	0.29	0.63	0.38	0.15	0.16	00.17

TABLE II

COMPARISON ON THE NUMBER OF PARAMETERS AND FLOPS

Datasets	Complexity	CDCNN	SSRN	HResNet	CEGCN	DMVL	S2FL
Salinas Valley	Parameters (M)	1.08	0.37	0.34	0.17	24.62	33.38
	FLOPs (M)	3.59	96.94	18.83	8566.24	1054.19	664.42
Pavia University	Parameters (M)	0.63	0.22	0.17	0.15	24.62	27.08
	FLOPs (M)	1.90	48.60	13.27	13079.92	1054.19	544.88
Trento	Parameters (M)	0.46	0.16	0.11	0.15	24.62	34.96
	FLOPs (M)	1.28	30.24	5.96	5727.40	1054.19	722.62
Houston 2018	Parameters (M)	0.40	0.14	0.09	0.13	24.62	33.51
	FLOPs (M)	1.03	23.47	6.39	86534.26	1054.19	669.11

the number of model parameters and the floating-point operations (FLOPs) to measure space and time complexity, respectively. The quantitative comparative results of model complexity for the S2FL model and several comparative methods are reported in Table II. It can be observed that supervised learning models generally have fewer parameters and FLOPs. Since the graph neural network [i.e., CNN-enhanced GCN (CEGCN)] employs the whole image to construct the adjacency matrix, its corresponding FLOPs are very large. Because the SSL models are pretrained on a large number of unlabeled samples, deep network models are generally employed to extract high-level features, thus having more parameters and FLOPs than supervised classification models. The SSL model adopts two-stage learning scheme, and its S2FL stage needs long training time, but the classification phase takes very little time.

C. Comparison With State of the Arts

In this section, we evaluate the proposed S2FL model from both quantitative and qualitative perspectives. The compared approaches include supervised, semi-supervised, and S2FL approaches. The supervised classification methods include the SVM [53], the contextual deep CNN (CDCNN) [10], the spectral–spatial unified network (SSUN) [54], the spectral–spatial residual network (SSRN) [11], and the hierarchical residual network (HResNet) [55]. Semi-supervised classification methods include the traditional transductive SVM (TSVM) [56] and the CEGCN [57]. Also,

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TABLE III

QUANTITATIVE RESULTS OF DIFFERENT APPROACHES ON THE SALINAS VALLEY SCENE. ONLY FIVE ANNOTATED SAMPLES PER CLASS ARE EMPLOYED FOR CLASSIFICATION. THE BEST RESULTS FOR EACH CATEGORY ARE SHOWN IN BOLD

Class No.	Supervised					Semi-Supervised		Self-Supervised			S2FL
	SVM	CDCNN	SSUN	SSRN	HResNet	TSVM	CEGCN	3DCAE	DFSL	DMVL	
C1	97.87	84.27	66.37	99.76	98.64	99.02	**99.89**	94.19	97.06	99.81	99.36
C2	98.74	59.81	81.27	99.56	99.82	99.62	**99.94**	97.75	98.67	99.82	99.77
C3	81.52	73.49	83.87	85.82	92.68	82.89	98.79	87.13	90.03	**99.13**	98.25
C4	97.30	75.35	99.65	93.66	95.38	97.42	**99.79**	89.57	92.24	70.90	95.60
C5	95.54	90.82	97.98	**99.44**	90.82	93.44	95.59	86.52	82.74	86.91	93.72
C6	99.62	95.53	98.36	99.69	99.92	99.86	99.16	**99.98**	99.35	95.19	99.88
C7	94.50	87.31	85.98	99.40	98.74	95.14	99.40	89.65	91.56	94.52	**99.43**
C8	68.83	67.22	44.40	73.01	78.93	86.34	77.43	77.64	65.33	97.94	**99.46**
C9	98.19	88.65	**99.98**	98.97	98.33	97.82	98.94	97.67	92.58	97.42	99.83
C10	79.66	83.89	94.16	**97.21**	91.06	85.17	87.52	81.92	79.42	96.70	93.08
C11	75.50	71.48	96.33	94.69	88.49	69.89	**98.63**	57.02	86.40	97.08	95.94
C12	86.06	74.60	98.92	96.96	96.46	89.98	99.31	86.76	95.80	95.90	**99.39**
C13	84.51	58.46	98.35	99.01	96.73	95.24	99.84	64.72	97.13	90.35	**99.94**
C14	77.51	69.56	96.15	**99.33**	94.54	96.88	95.49	88.14	77.48	71.87	90.91
C15	51.63	56.51	87.99	55.11	54.33	75.35	78.27	56.08	67.49	**99.36**	93.96
C16	80.01	57.49	91.29	99.86	99.18	87.35	96.42	83.95	87.01	99.64	**99.98**
OA	80.28	76.02	81.70	86.65	85.07	86.21	91.03	81.07	88.05	94.85	**97.66**
	0.21	0.71	0.37	0.45	0.35	0.29	0.15	0.19	0.25	0.53	**0.17**
AA	85.44	74.65	88.82	93.22	92.13	90.71	95.28	83.67	87.52	93.28	**97.41**
	0.15	0.85	0.51	0.38	0.39	0.38	0.32	0.16	0.29	0.69	**0.16**
κ	78.17	72.38	79.81	85.16	83.43	84.49	90.04	79.07	87.21	94.27	**97.40**
	0.23	0.78	0.42	0.40	0.38	0.32	0.17	0.20	0.26	0.59	**0.22**

TABLE IV

QUANTITATIVE RESULTS OF DIFFERENT APPROACHES ON THE PAVIA UNIVERSITY SCENE. ONLY FIVE ANNOTATED SAMPLES PER CLASS ARE EMPLOYED FOR CLASSIFICATION. THE BEST RESULTS FOR EACH CATEGORY ARE SHOWN IN BOLD

Class No.	Supervised					Semi-Supervised		Self-Supervised			S2FL
	SVM	CDCNN	SSUN	SSRN	HResNet	TSVM	CEGCN	3DCAE	DFSL	DMVL	
C1	90.78	94.92	70.99	99.33	**99.61**	95.42	95.71	85.17	80.31	69.02	97.70
C2	78.72	87.92	73.21	95.11	95.36	82.08	95.26	81.76	90.26	90.71	**98.60**
C3	34.21	37.02	55.64	39.67	67.86	52.65	59.99	59.11	85.65	87.72	**89.93**
C4	55.24	48.88	**96.40**	89.36	92.02	77.54	70.92	75.26	67.66	92.74	88.98
C5	90.25	93.87	98.43	99.77	99.85	85.93	**99.95**	97.42	73.17	86.21	97.26
C6	31.28	37.04	64.01	47.67	65.69	38.43	39.76	36.35	57.25	89.99	**97.24**
C7	37.94	31.33	89.58	42.12	61.82	50.71	84.96	39.24	65.89	86.92	**97.59**
C8	65.81	65.51	91.84	66.51	81.24	77.75	**93.77**	77.49	56.77	90.46	87.28
C9	99.76	57.29	99.26	99.89	**99.93**	99.92	99.92	89.11	93.39	69.37	99.75
OA	58.69	58.34	76.06	76.05	86.86	70.09	85.13	69.28	80.31	85.88	**96.00**
	0.63	0.62	0.41	0.28	0.23	0.22	0.13	0.10	0.51	0.14	**0.24**
AA	64.89	61.53	82.15	75.49	84.82	73.38	82.25	71.21	74.48	84.79	**94.93**
	0.82	0.70	0.56	0.36	0.31	0.25	0.21	0.14	0.34	0.28	**0.23**
κ	49.17	50.32	69.55	69.96	82.96	62.11	79.68	60.55	76.93	81.19	**94.72**
	0.63	0.61	0.39	0.31	0.25	0.19	0.17	0.18	0.58	0.18	**0.21**

TABLE V

QUANTITATIVE RESULTS OF DIFFERENT APPROACHES ON THE TRENTO SCENE. ONLY FIVE ANNOTATED SAMPLES PER CLASS ARE EMPLOYED FOR CLASSIFICATION. THE BEST RESULTS FOR EACH CATEGORY ARE SHOWN IN BOLD

Class No.	Supervised					Semi-Supervised		Self-Supervised			S2FL
	SVM	CDCNN	SSUN	SSRN	HResNet	TSVM	CEGCN	3DCAE	DFSL	DMVL	
C1	49.97	88.01	87.08	92.37	87.02	82.40	**98.90**	85.12	78.89	91.83	98.53
C2	83.77	73.74	85.03	85.23	**87.95**	73.58	84.99	80.23	83.35	83.44	79.93
C3	70.85	65.39	80.88	50.32	71.71	74.74	**98.95**	74.63	26.86	78.94	92.41
C4	99.22	91.81	90.08	98.65	99.54	99.16	98.58	99.34	99.81	**99.84**	99.05
C5	87.87	91.86	97.36	95.26	97.42	96.03	95.65	94.83	95.75	97.24	**99.83**
C6	80.98	84.51	41.37	87.12	80.85	73.46	57.96	73.11	69.22	86.46	**89.82**
OA	80.13	87.33	86.47	93.19	91.75	88.81	92.04	90.71	87.04	94.31	**96.26**
	0.11	0.43	0.26	0.19	0.29	0.20	0.39	0.42	0.25	0.45	**0.13**
AA	78.78	82.55	80.30	84.83	87.42	83.23	89.17	84.54	75.65	89.63	**93.26**
	0.22	0.56	0.32	0.22	0.48	0.38	0.47	0.57	0.27	0.76	**0.21**
κ	74.19	82.85	82.04	90.94	90.39	85.20	89.42	87.63	82.97	92.43	**95.00**
	0.14	0.60	0.30	0.17	0.37	0.26	0.35	0.48	0.24	0.60	**0.17**

the S2FL methods are the 3-D convolutional autoencoder (3DCAE) [33], the deep few-shot learning (DFSL) model [25], and the deep multiview learning (DMVL) model [37].

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TABLE VI

QUANTITATIVE RESULTS OF DIFFERENT APPROACHES ON THE HOUSTON 2018 SCENE. ONLY 20 ANNOTATED SAMPLES PER CLASS ARE EMPLOYED FOR CLASSIFICATION. THE BEST RESULTS FOR EACH CATEGORY ARE SHOWN IN BOLD

Class No.	Supervised					Semi-Supervised		Self-Supervised			S2FL
	SVM	CDCNN	SSUN	SSRN	HResNet	TSVM	CEGCN	3DCAE	DFSL	DMVL	
C1	52.86	34.29	85.75	66.16	49.91	63.57	84.76	**96.10**	64.90	26.97	68.44
C2	78.24	80.36	68.54	83.30	69.11	81.76	76.49	**83.50**	82.07	72.91	76.10
C3	11.57	2.61	97.46	31.99	49.33	6.49	28.95	19.70	10.31	11.72	**99.42**
C4	80.80	72.24	82.25	81.32	75.51	73.85	**83.23**	78.23	82.90	60.04	79.07
C5	23.23	29.48	61.63	37.00	21.88	17.02	30.56	**73.19**	15.40	17.97	20.91
C6	17.42	12.80	**98.44**	20.20	24.20	25.57	87.94	80.72	56.13	65.26	47.57
C7	49.40	1.97	9.20	19.83	56.87	9.39	25.53	19.59	2.40	2.52	**91.07**
C8	48.62	58.69	45.09	53.24	49.91	51.45	67.47	49.26	70.30	**96.41**	61.20
C9	95.59	97.03	33.27	97.49	**98.23**	94.33	54.75	32.35	94.06	96.50	96.53
C10	32.11	25.20	41.73	42.66	42.27	37.42	33.92	19.23	48.79	50.76	**53.57**
C11	29.46	43.42	17.25	37.97	39.38	35.58	38.19	19.44	33.06	23.06	**49.56**
C12	1.96	2.93	7.38	2.72	1.96	2.27	4.98	**31.89**	2.32	6.04	3.82
C13	36.61	38.25	43.37	43.41	53.74	42.51	53.62	40.92	52.57	61.54	**62.49**
C14	38.31	39.81	**79.17**	44.80	34.70	34.49	61.78	70.99	49.63	58.72	59.08
C15	33.54	28.19	84.34	58.41	27.37	47.56	55.43	84.05	65.26	**90.96**	88.00
C16	39.80	31.18	**85.38**	51.16	32.09	31.26	64.89	64.51	63.11	82.25	79.70
C17	2.42	1.23	9.43	4.01	1.09	3.24	10.00	**19.22**	1.46	1.94	7.48
C18	14.53	15.64	61.86	21.68	14.89	23.81	65.19	61.09	34.76	43.98	**82.66**
C19	16.09	12.50	56.77	28.01	19.32	31.84	68.89	63.09	71.35	89.68	**92.17**
C20	45.07	52.91	**87.68**	49.83	32.79	24.43	86.55	84.05	66.27	71.36	44.22
OA	43.60	40.95	45.02	52.73	44.14	47.01	59.14	42.74	58.10	63.11	**69.63**
	0.27	0.41	0.56	0.22	0.48	0.38	0.26	0.42	0.26	0.16	**0.13**
AA	37.38	34.04	55.65	43.76	39.73	36.89	55.16	54.56	48.35	51.53	**63.15**
	0.15	0.48	0.62	0.19	0.34	0.35	0.30	0.49	0.24	0.18	**0.22**
κ	37.20	34.87	38.55	46.28	38.13	40.34	52.16	36.10	52.12	55.89	**63.03**
	0.22	0.39	0.49	0.22	0.45	0.34	0.22	0.37	0.19	0.16	**0.14**

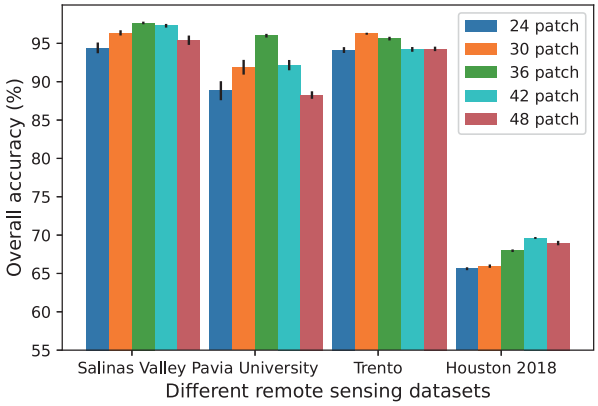

Fig. 10. Sensitivity analysis of patch size on four benchmark datasets.

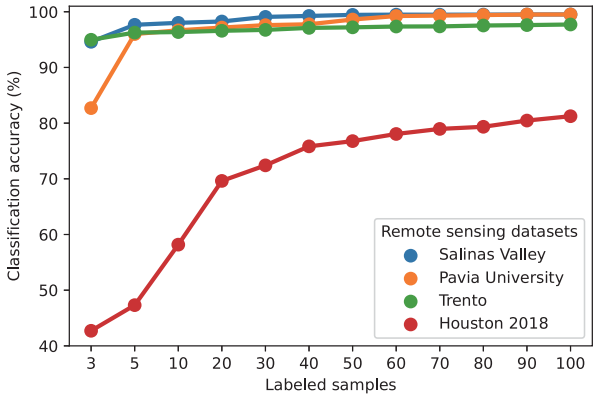

Fig. 11. Classification accuracies with different numbers of labeled samples on four benchmark datasets.

More specifically, we adopt widely used classification evaluation coefficients, i.e., overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and individual category accuracy, to evaluate the classification performance quantitatively. The feature visualization and resulting classification maps are also employed to appraise the feature learning and classification performance from visual perspective.

1) Quantitative Evaluation: Tables III–VI report main evaluation coefficients and individual category accuracy of different approaches on four benchmark datasets. To increase the reliability of experimental results, we conduct ten trials for each classification experiment.

Despite the remarkable success of supervised learning paradigm, when employing a few labeled samples for land cover classification, the supervised classification approaches cannot achieve good classification results. This indicates that when there are insufficient annotated samples, supervised learning methods encounter the overfitting phenomenon, thus seriously affecting the classification performance. Utilizing both labeled and unlabeled samples, semi-supervised classification methods can learn more discriminative features. As a result, semi-supervised methods (i.e., TSVM and CEGCN) generally have higher classification accuracies than supervised ones (e.g., CDCNN and SSUN).

As a novel classification paradigm, the SSL methods learn key discriminative features from unlabeled data, which effectively improve the land cover classification performance. Based on the convolutional autoencoder, the 3DCAE extracts spatial–spectral features in an unsupervised way, but its classification performance is also limited when using a few labeled samples for classification. The other two feature learning-based methods (i.e., DFSL and DVML) have better classi-

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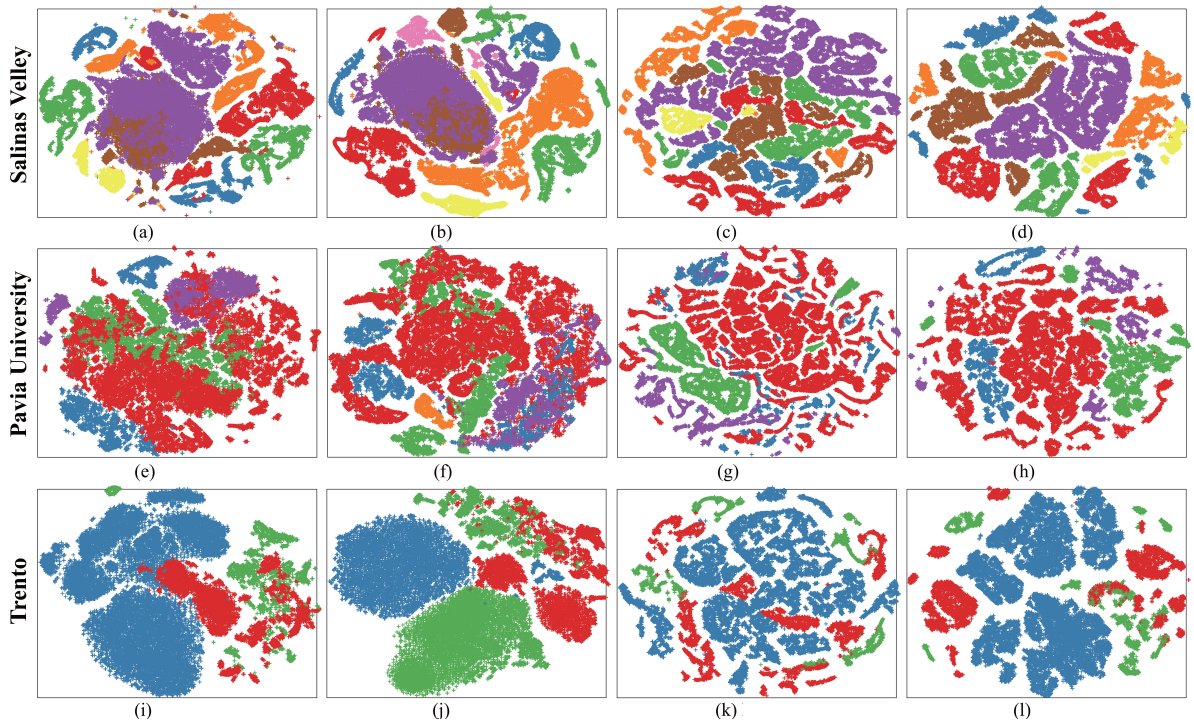

Fig. 12. t-SNE visualization of features obtained by different models on three benchmark datasets. The Salinas Valley dataset—(a) Original spectral, (b) DFSL, (c) DMVL, and (d) S2FL. The Pavia University dataset—(e) Original spectral, (f) DFSL, (g) DMVL, and (h) S2FL. The Trento dataset—(i) Original spectral, (j) DFSL, (k) DMVL, and (l) S2FL.

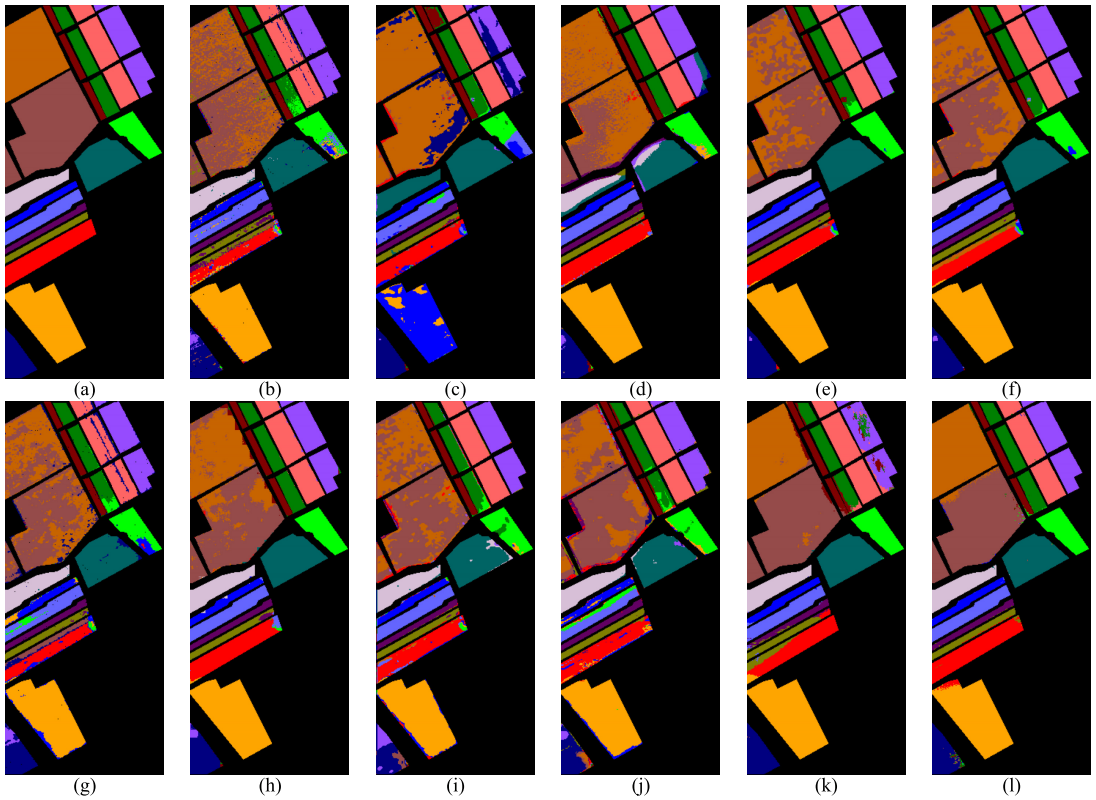

Fig. 13. Ground-truth and classification maps on the Salinas Valley dataset. (a) Ground truth. (b) SVM (OA: 80.28%). (c) CDCNN (OA: 76.02%). (d) SSUN (OA: 81.70%). (e) SSRN (OA: 86.65%). (f) HResNet (OA: 85.07%). (g) TSVM (OA: 86.21%). (h) CEGCN (OA: 91.03%). (i) 3DCAE (OA: 81.07%). (j) DFSL (OA: 88.05%). (k) DMVL (OA: 94.85%). (l) S2FL (OA: 97.66%).

fication performance, which means that they have learned more discriminative features for classification. Especially, the DMVL, based on the contrastive learning paradigm, can lean meaningful features from unlabeled data, and the corresponding classification accuracies are higher than other comparative methods.

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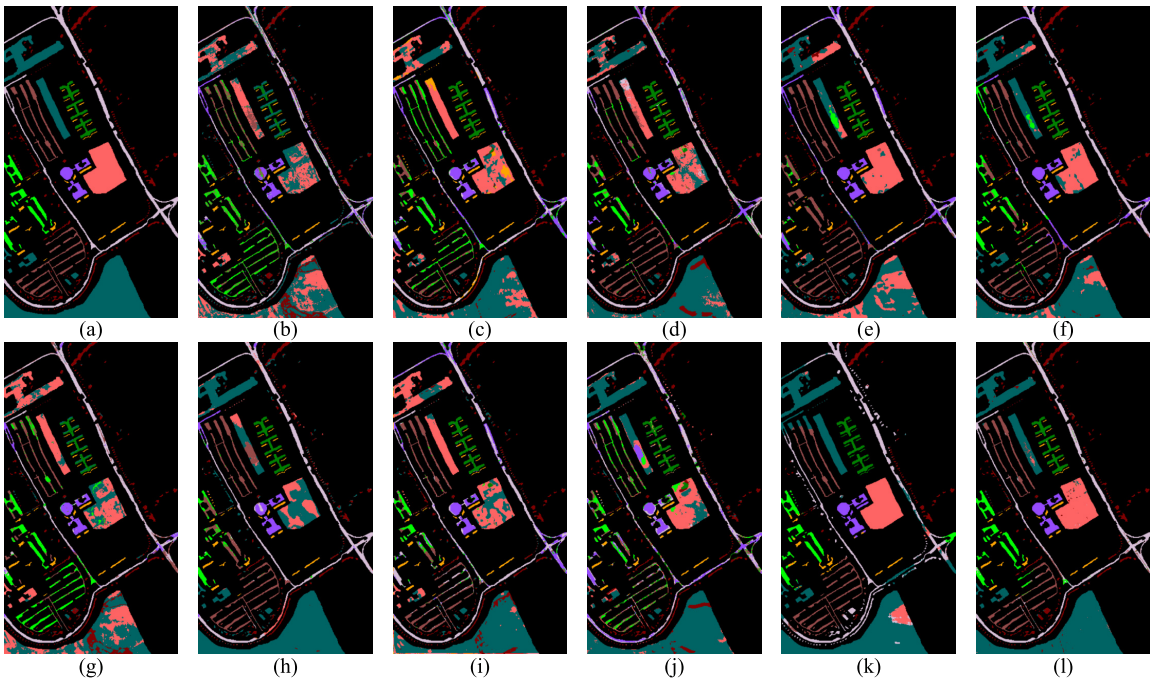

Fig. 14. Ground-truth and classification maps on the Pavia University dataset. (a) Ground truth. (b) SVM (OA: 58.69%). (c) CDCNN (OA: 58.34%). (d) SSUN (OA: 76.06%). (e) SSRN (OA: 76.05%). (f) HResNet (OA: 86.86%). (g) TSVM (OA: 70.09%). (h) CEGCN (OA: 85.13%). (i) 3DCAE (OA: 69.28%). (j) DFSL (OA: 80.31%). (k) DMVL (OA: 85.88%). (l) S2FL (OA: 96.00%).

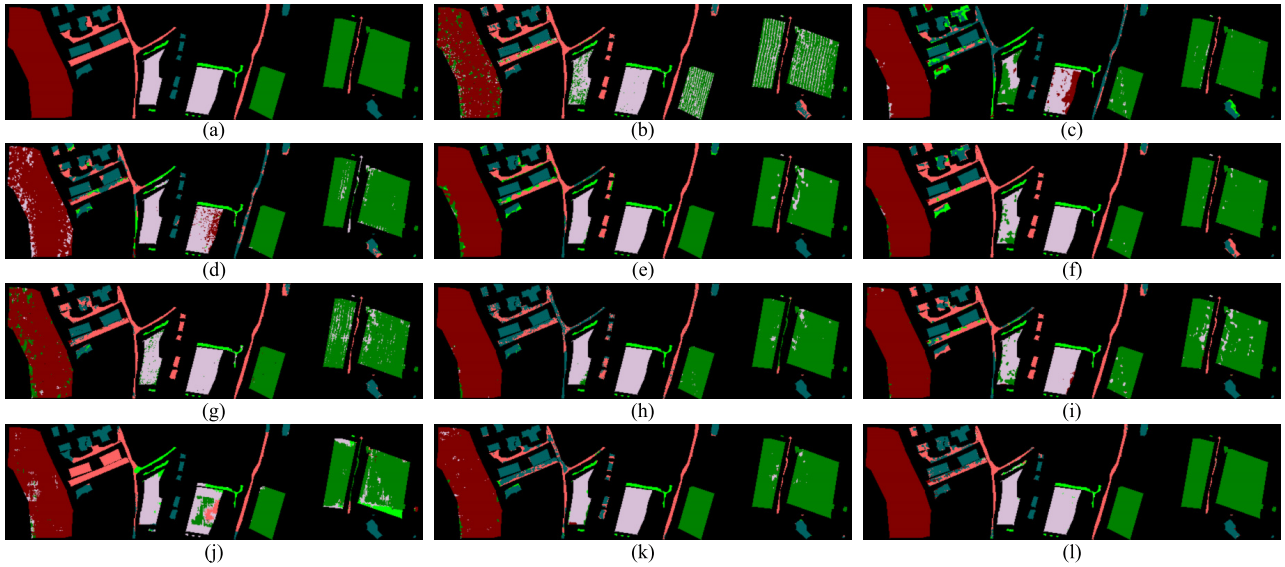

Fig. 15. Ground-truth and classification maps on the Trento dataset. (a) Ground truth. (b) SVM (OA: 80.13%). (c) CDCNN (OA: 87.33%). (d) SSUN (OA: 86.47%). (e) SSRN (OA: 93.19%). (f) HResNet (OA: 91.75%). (g) TSVM (OA: 88.81%). (h) CEGCN (OA: 92.04%). (i) 3DCAE (OA: 90.71%). (j) DFSL (OA: 87.04%). (k) DMVL (OA: 94.31%). (l) S2FL (OA: 96.26%).

Furthermore, the investigated generative S2FL method is an effective S2FL pattern. Using the same number of annotated samples, the investigated approach achieves the best classification accuracy in terms of main evaluation coefficients. The reason is that the S2FL model utilizes multiple observed views constructed from multimodal RS images for generative SSL, thereby effectively learning key features for classification task, which provides a feasible solution to relieve the urgent need for manually labeled samples.

2) Feature Visualization: Apart from the abovementioned quantitative analysis, the distributed stochastic neighbor embedding (t-SNE) algorithm [58] is employed to display the distribution of learned features in the 2-D space. The feature distribution maps obtained by different feature learning approaches on the Salinas Valley, Pavia University, and Trento datasets are shown in Fig. 12, in which different colors represent different classes. In Fig. 12(a), (e), and (i), we can observe that the original spectral features of different categories are twined, and the discriminant capability is poor. The transfer learning-based DFSL and the DMVL based on contrastive learning increase the distances between different categories, which indicates that the separability of different

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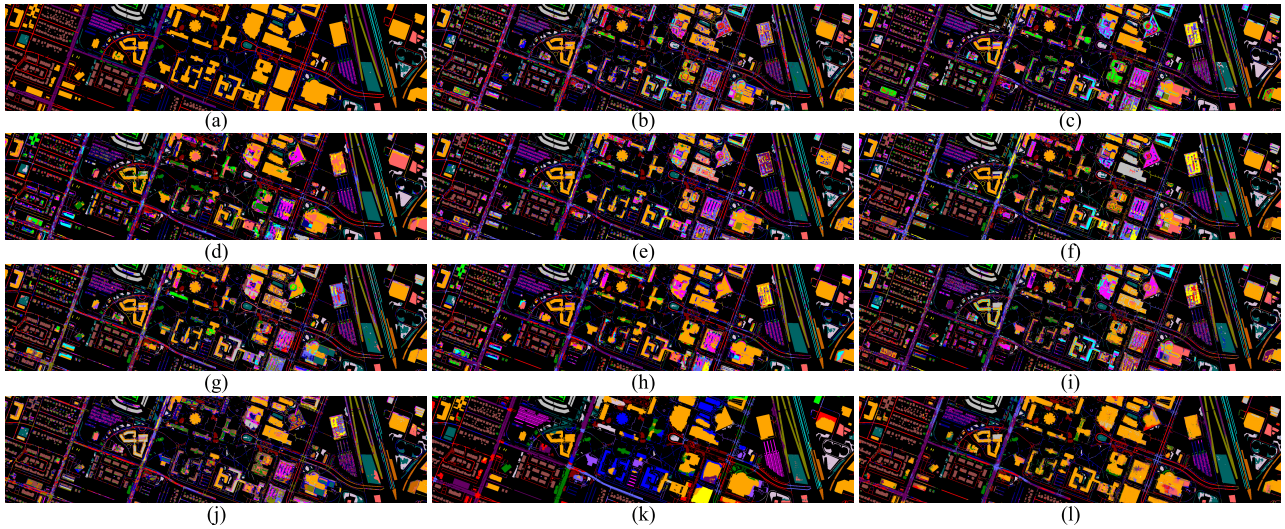

Fig. 16. Ground-truth and classification maps on the Houston 2018 dataset. (a) Ground truth. (b) SVM (OA: 43.60%). (c) CDCNN (OA: 40.95%). (d) SSUN (OA: 45.02%). (e) SSRN (OA: 52.73%). (f) HResNet (OA: 44.14 %). (g) TSVM (OA: 47.01%). (h) CEGCN (OA: 59.14%). (i) 3DCAE (OA: 42.74%). (j) DFSL (OA: 58.10%). (k) DMVL (OA: 63.11%). (l) S2FL (OA: 69.63%).

TABLE VII
ABLATION STUDY WITH DIFFERENT MODAL FEATURES ON FOUR BENCHMARK DATASETS

	Salinas Valley			Pavia University			Trento			Houston 2018		
	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
Spectral	80.28	85.44	78.17	58.69	64.89	49.17	80.13	78.78	74.19	43.60	37.38	37.20
	0.21	0.15	0.23	0.63	0.82	0.63	0.11	0.22	0.14	0.27	0.15	0.22
Spatial	88.96	86.48	87.81	80.06	72.58	73.90	88.12	73.73	84.21	56.58	49.43	48.78
	0.19	0.13	0.16	0.59	0.74	0.76	0.46	0.50	0.45	0.49	0.55	0.47
S2FL	97.66	97.41	97.40	96.00	94.93	94.72	96.26	93.26	95.00	69.63	63.15	63.03
	0.17	0.16	0.22	0.24	0.23	0.21	0.13	0.21	0.17	0.13	0.22	0.14

classes has been increased. As for Fig. 12(d), (h), and (l), the samples of the same category are clustered together, and the samples of different classes are separated away, which means that the S2FL model can extract more discriminative features compared with other approaches.

3) Classification Maps: The resulting classification maps are also employed for visual evaluation. Classification maps obtained by different approaches on four benchmark RS datasets are shown in Figs. 13–16, respectively. In Figs. 13–16, land cover categories are represented by different colors, and ground-truth maps are also displayed for better visual comparison. It can be observed from Figs. 13–16 that there are less noisy pixels in classification maps obtained by semi-supervised classification models and unsupervised feature extraction methods. Also, the classification maps obtained by the S2FL method are more homogeneous, which have the fewest noisy pixels. This is because the proposed model could extract more discriminative features using multimodal RS images for generative S2FL. Also, the lightweight classifier achieves better classification performance using learned features.

D. Ablation Analysis

1) Different Modal Features: Based on the fact that there are multimodal features contained in RS data, the proposed S2FL model aims to learn multimodal features for land cover classification. In this section, we analyze the effects of different modal features on the classification performance. Since spectral and spatial characteristics simultaneously contribute to land cover recognition, we utilize the original spectral features, self-supervised learned spatial features, and fused spectral–spatial features for classification. The classification results using different modal features are reported in Table VII. From Table VII, we can observe that the discriminative capability of single modal feature is limited, and the corresponding classification accuracies are generally low. By integrating the learned spatial features with the corresponding spectral information, the fused features are more conductive to land cover classification task.

2) Self-Supervised Feature Learning: Because the SSL algorithm is mainly conducted in the spatial domain, we also analyze the impact of this SSL on classification performance. For vanilla spatial information, we take the information of local images as feature vector and compare it with the spatial features after SSL, and comparative results are reported in Table VIII. Table VIII indicates that the proposed self-supervised algorithm is effective, which improves the land cover classification performance significantly, especially in the small sample classification case.

3) Different Source Data: Because the proposed approach aims to extract features from multimodal RS data for classification task, we analyze the effectiveness of this method in handling multimodal data and the impact of different source data on classification results. There are HSI and other type

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TABLE VIII
ABLATION STUDY OF THE SSL ON FOUR BENCHMARK DATASETS

	Salinas Valley			Pavia University			Trento			Houston 2018		
	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
vanilla	80.86	83.85	78.80	63.63	67.01	54.20	80.13	78.78	74.19	44.20	36.09	37.34
	0.15	0.12	0.17	0.26	0.11	0.19	0.11	0.22	0.14	0.12	0.16	0.11
self-supervised	97.66	97.41	97.40	96.00	94.93	94.72	96.26	93.26	95.00	69.63	63.15	63.03
	0.17	0.16	0.22	0.24	0.23	0.21	0.13	0.21	0.17	0.13	0.22	0.14

TABLE IX
ABLATION STUDY OF DIFFERENT SOURCE DATA ON CLASSIFICATION PERFORMANCE

	Trento			Houston 2018		
	OA	AA	κ	OA	AA	κ
HSI	94.51	86.16	92.68	63.68	58.36	56.83
	0.41	0.63	0.54	0.44	0.49	0.41
Multisource	96.26	93.26	95.00	69.63	63.15	63.03
	0.13	0.21	0.17	0.13	0.22	0.14

of RS data in Trento and Houston 2018 datasets, and we employ hyperspectral data and multimodal RS images for comparative experiments, respectively. The experiment results using different source data are reported in Table IX. It can be observed that using multimodal data for feature learning can effectively improve classification accuracy, which also indicates that the proposed feature learning method is an effective multimodal SSL model.

V. CONCLUSION

In this study, we investigate a generative SSL architecture for multimodal RS image feature learning and land cover classification. Based on the fact that there are multimodal RS data within the same observed scenes, we put forward to construct multiple complementary views from RS data for feature learning. Then, we propose an effective generative SSL scheme to learn key feature representations from multiview data. In addition, to make full use of the complementary information contained in multimodal data, we further normalize the learned features and combine them with the corresponding spectral information. We carry out four groups of benchmark datasets for feature learning and land cover classification, and ablation studies as well as comparative experiments have confirmed the effectiveness and superiority of the investigated model. In this research, we only employ the lightweight SVM for classification, and we will explore more classifiers for fine-tuning in our future research work. Another research direction is the dynamic fusion of multimodal features in the SSL scheme.

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